

Stochastic Calculus Workshop / Buyout Case

Session 25 • Track 1: Buyout case workshop • Track 2: Full OU process derivation

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 25 of 32

Session 25: split-track agenda

TRACK 1 (Practitioner)

Buyout case workshop

- ▶ Case: KKR/RJR Nabisco — apply GE-LAV to historical buyout
- ▶ Calibrate $L(t)$ over 1989-1995 period
- ▶ Compute IRR, LA-IRR, PME, LA-PME
- ▶ Group exercise: would GE-LAV have changed the deal?
- ▶ Deliverable: 2-page case memo

TRACK 2 (Researcher)

Full OU process derivation

- ▶ Existence and uniqueness of OU SDE solution
- ▶ Derivation via Itô applied to integrating factor $\exp(\kappa t) \cdot L(t)$
- ▶ Stationary distribution proof: $\text{Normal}(\bar{L}, \sigma^2/2\kappa)$
- ▶ Karatzas-Shreve Section 5.6 walk-through
- ▶ Problem: derive transition density $p(L_t | L_s)$

Session 25 summary

What we accomplished today

- 1 Session 25 runs split-track: practitioners and researchers work on different deliverables
- 2 Track 1: case-based, applied analysis with deliverable IC memo
- 3 Track 2: theoretical, with deliverable problem set or proof
- 4 Both tracks reconvene in Session 32 for final project presentations

Next session

Session 26: continued split-track or capstone